BIOBUZZ (2026-27) Competition Manual Digest

Team notes from a full read of the **V1 Competition Manual** (published at kickoff, September 2026). Source of truth is the manual itself: <https://ftc-resources.firstinspires.org/ftc/game/manual>. Team Updates post every **Thursday**; they change this document. The Q&A opens **Sept 28, 2026**.

Where this digest gives an opinion for our robot, it is marked **[our take]**.

1. The game in one paragraph

Two alliances of two robots. Each match is **30 s AUTO, 8 s hands-off transition, 2:00 TELEOP**. Robots collect two kinds of balls, POLLEN and NECTAR, and score them three ways: **LAUNCH** them into the upward-facing CELL of their HIVE to make it **TIP** (20 pts each tip), **place** them into the top of FLOWERS in the last minute (2 pts each, only for the alliance that "owns" the flower), and push them into their **GARDEN** corner (1 pt each). Robots also score for leaving the wall in AUTO and parking in the LOADING ZONE.

2. Game pieces (section 9.8)

Piece	Size	Weight	Count per match	Where they start
POLLEN, yellow	2.8 in / 71 mm	~25 g	40	4 in each of 4 FLOWERS (16), 4 in each GARDEN (8), 4 preloaded on each robot (16)
NECTAR, red or blue	3.6 in / 91 mm	~41 g	8 per alliance	3 in each alliance's upward CELL (6), 5 held by each alliance's humans (10)

Both are Gopher ResisDent polyethylene balls, "not perfectly spherical and may vary in size." Design for that.

3. Field (section 9)

- 12 ft × 12 ft, 36 soft foam tiles, standard AndyMark perimeter. FIRST Championship fields sit on 24 in risers.
- **HIVE Structure** in the center: a frame 49.5 in wide × 39 in deep at the base. Two HIVES (one red, one blue) each pivot on an axis **43.95 in above the tiles**. Each HIVE has two CELLS 18.8 in apart; only one faces up at a time. A CELL opening is about **20 in wide × 14 in tall × 12 in deep**. The HIVE is bi-stable and tips when enough balls land in the upward cell.
- **AprilTags** (36h11, 3.25 in) are on the underside of every CELL, facing down toward the tiles: IDs 30-33 and 34-37 on the red cells, 38-41 and 42-45 on the blue cells.

- **FLOWERS**: 4 of them, on the perimeter walls. Top opening is a **4 in diameter ring, 21.5 in above the tiles**, with a 1.25 in backstop. A retrieval opening at the bottom (3.55 in tall × 3.57 in deep) lets robots pull POLLEN out, but NECTAR cannot come out.
- **LOADING ZONE**: 23 in × 11 in against the wall by each alliance area. Humans enter NECTAR here; robots PARK here.
- **GARDEN**: a 23 in × 2 in tape strip in opposite corners, one per alliance. Not a protected zone.
- **ALLIANCE AREA**: 97 in × 54 in outside the field where the drive team stands.

4. Scoring (Table 10-2)

Achievement	AUTO	TELEOP
LEAVE (robot no longer touching perimeter wall)	3	
PARK (robot at least partially in LOADING ZONE)	5	5
HIVE TIP	20	20
Each POLLEN or NECTAR left in an upward CELL at match end		2
Bottom NECTAR bonus (lowest scoring nectar in a flower is yours)		5
Each ball in a FLOWER you own		2
Each ball in your GARDEN		1

Ranking points (qualification only): WIN 3, TIE 1, **SWARM RP** for LEAVE + PARK points at or above a threshold (16 at regular events), **POLLINATOR 1** for at least 4 tips, **POLLINATOR 2** for at least 7 tips. Championship thresholds are TBA. Ranking is by average RP, then average match points minus fouls, then average tips, then average AUTO points.

How the mechanics chain together

1. Every HIVE TIP unlocks **one NECTAR** that a human may throw into the LOADING ZONE (G426). With **60 s left**, all remaining NECTAR may be entered.
2. Humans enter NECTAR by hand only, and it must **touch the tile inside the LOADING ZONE before touching anything else** (G427).
3. NECTAR may only go into a FLOWER in the **last 60 seconds** (G410, major foul per nectar). The manual says flower scoring starts at one minute; earlier placements "are still scored but subject to penalties." Watch the Q&A for whether early POLLEN is penalized.
4. The alliance whose NECTAR is **top-most** in a flower **owns** it and gets 2 pts for every ball in it, no matter who placed them. Lowest NECTAR of your color earns the 5 pt bonus.
5. Balls that spill from a tipped HIVE must hit something before a robot may collect them (G409).

5. Gameplay rules that shape the robot

- **G407: never CONTROL more than 4 balls at once.** Momentary 5th is tolerated, 6 is strategic. Guards to stop balls piling on top of the robot are explicitly encouraged.
- **G408:** don't control the opponent's NECTAR.
- **G417:** the only legal way to move a HIVE is launching into the upward cell. No ramming, no launching at the outside of a cell.
- **G418:** POLLEN and NECTAR go in the top of a FLOWER only; only POLLEN comes out, only through the bottom. Contacting the flower while scoring is fine.
- **G405:** don't deliberately eject balls from the field. **G406:** don't gouge or mark the balls.
- **G411:** no hoarding or blocking access to balls. Gardens are not protected.
- **G421:** 3 second pin count. **G419/G420:** no damage, tipping, or entangling. Expect a "highly interactive," high-contact game. Exposed intakes and wiring get no protection from the rules.
- **G402:** stay on your own half in AUTO (columns A-C red, D-F blue).
- **G403/G404:** no powered motion during the 8 s transition or after the buzzer.

AUTO start (G304)

Robot starts touching the perimeter wall on its own side, not in the LOADING ZONE, not touching a FLOWER, inside the 18 in cube, **touching exactly 4 preloaded POLLEN** (on the robot or on the tiles touching it). An OpMode must be INIT'd before the match (G305), even if there's no autonomous.

6. Robot construction rules (section 12)

Rule	Limit
R102	**18 × 18 × 18 in** starting configuration. Preloads may stick out.
R105	Expansion limit **18 × 24 in footprint, 29 in tall**, at all times, enforced mechanically, not in software.
R104	No weight limit.
R503	**8 motors and 8 servos** total across all configurations.
R501	goBILDA 5201/5202/5203/5204 and 5000 series are legal, plus REV, AndyMark, Studica, Tetrix, etc.
R502	Servos: ≤ 8 W and ≤ 4 A stall at 6 V. Linear servos ≤ 1 A.
R601	Exactly one 12 V NiMH battery from the approved list (goBILDA 3100-0012-0020 is on it).
R603	One approved main power switch (goBILDA Floodgate is legal).

Rule	Limit
R701	One REV Control Hub (or phone + Expansion Hub), plus at most one Expansion Hub.
R707/R708	USB is for UVC webcams and the **Limelight 3A** only. No stereo cameras. Limelight 3G, OAK, OpenMV are banned.
R704.D	**No FTC Dashboard or other streaming tools during matches.** Only the Driver Station app may carry telemetry. No video streams.
R705	Name devices `<team>-RC` and `<team>-DS`.
R711	Change the Control Hub Wi-Fi password. Bluetooth off.
R801	No pneumatics, blowers, vacuums, or fans that move balls. Flywheels and rollers are fine.
R301/R303	No COTS mechanism that solves a game task by itself, except drive chassis and StarterBot parts. COTS mechanisms must be single degree of freedom.
R305	Can't build the robot out of game pieces or replicas.
R201	Nothing that damages tiles or sheds material. Sharp edges and abrasive surfaces get inspected.

Robot signs (R401-R403): at least two, on faces ≥ 90° apart, each with a solid red or blue rectangle at least 6.5 × 2.5 in and white team numbers about 2.25 in tall, not vertically stacked, not illuminated. Must be readable from 12 ft.

7. People and event rules

- **Drive team:** up to 4. One may be an adult (the DRIVE COACH). Drivers and the HUMAN PLAYER are students. Badges above the waist.
- Coaches may hold the Driver Station and press INIT/START/STOP, but may not touch gamepads (G423) or balls (G424).
- Safety glasses in the pits and arena. Closed shoes. No team Wi-Fi hotspots or Bluetooth anywhere in the venue (E301).
- Battery chargers: 3 A per channel max, proper polarized connectors only (E511).
- Pits: small benchtop tools only, no grinders or torches, soldering with an electric iron is fine.
- Only one robot may be inspected and played per event (E117).
- Re-inspection is required after most changes except code, fasteners, signs, and identical part swaps (I303).

8. Tournament format (section 13)

- 5 or 6 qualification matches per team. Rankings by RANKING SCORE (average RP).

- Alliance selection, then **double-elimination** playoffs. 2 alliances for 4-10 teams, 4 for 11-20, 6 for 21-40, 8 for 41-64.
- Only students may talk to the Head Referee at the question box, max two people, and questions must be asked before alliance selection.
- No video review, ever.
- **Advancement points:** qualification rank (2-16 pts), alliance captain or draft order (21 minus position), playoff finish (40/20/10/5), judged awards (Inspire 60/30/15, other awards 12/6/3). Teams advance from their first three qualifying or league events, in their home region only.
- Registration cutoff that affects championship slot allocation: **Nov 17, 2026**.

9. Judging (section 6)

- **Portfolio:** cover page + **15 pages max**, US Letter or A4, under 15 MB if digital, content since **Jan 1, 2026** only. First names and last initials only. Judges won't click links.
- **Initial interview:** at least 10 minutes, first 5 for an uninterrupted presentation, at least 2 students, one silent adult observer allowed. Two standard questions come from a published question bank.
- Awards: Inspire (needs portfolio and strength in all three categories), Think (portfolio), Connect / Reach / Sustain (team attributes), Innovate / Control / Design (machine). Control Award **requires** a portfolio section on sensors and software. A team can win only one judged award per event.

10. Match timeline cheat sheet

Clock	What happens
2:30	Match starts. AUTO. No touching controllers.
2:00	AUTO ends, 3 buzzers. 8 s hands-off transition. Tips completed before TELEOP count as AUTO.
1:52	TELEOP begins.
1:00	FLOWER ownership unlocked. All remaining NECTAR may be entered. NECTAR may go into flowers.
0:20	Train whistle.
0:00	Robots stop. Cells, flowers, gardens, and TELEOP PARK are counted once everything settles.

11. What this means for our robot [our take]

- **Two ball sizes, 71 and 91 mm.** Every intake, hopper, and shooter path must pass a 91 mm ball with margin and still grip a 71 mm one.
- **Capacity is capped at 4.** Design the hopper for exactly 4 and add guards so a 5th can't ride on top. This also matches the 4 preloaded POLLEN.

- **A launcher is required to score the big points.** The HIVE pivot is at 44 in and the cell opening is above that; our robot may only be 29 in tall when extended. Nothing else earns 20 pts a shot, and 4 tips is a ranking point.
- **Flower scoring needs a lift and precision.** The 4 in opening at 21.5 in is only 0.4 in wider than a NECTAR. A ball dropped from above with a guide is realistic; a shot is not.
- **Pulling POLLEN out of a flower bottom** means a ground-level intake that can reach into a 3.55 in tall slot against the wall. Our intake spokes are pointed at exactly this.
- **AprilTags face downward** from the cells, so a camera looking up at the hive is how you localize for shots. Only UVC webcams or a Limelight 3A are legal.
- **Programmers: no FTC Dashboard at competition.** Keep tuning tools switchable and use Driver Station telemetry in matches.
- **Robust matters.** The manual repeats that this is a contact game and that fragile intakes are your own problem.
- **Auto priorities:** LEAVE (3) + PARK (5) gets most of the SWARM RP by itself; tipping the hive with the 4 preloads is the big auto prize.